

Milestone 1 Progress Evaluation

Project Information

Project Title :

Skin Cancer Detection App Using Machine Learning

Names and Email Addresses :

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Faculty Advisor :

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Client :

Dr. Zahra Nematzadeh, Department of Electrical Engineering and Computer Science, FIT

Progress of Current Milestone

Milestone 1 Progress Matrix :

Task	Completion %	Elif	Neva	Samuel	Sophia	To do
Compare and Select Technical Tools	100%	25%	25%	25%	25%	none
“Hello World” Demos	100%	25%	25%	25%	25%	none
Resolve Technical Challenges	100%	25%	25%	25%	25%	none
Compare and Select Collaboration Tools	100%	25%	25%	25%	25%	none
Requirements Document	100%	25 %	25%	25%	25%	none

Design Document	100%	25%	25%	25%	25%	none
Test Plan	100%	25%	25%	25%	25%	none

Discussion of Accomplished Tasks and Obstacles for the Current Milestone

Task 1: Compare and Select Technical Tools

The team compared and selected technical tools across the system. For image preprocessing, Sophia selected OpenCV and Python. She used Google Colab as the environment to run demos and compare tools for image preprocessing. Through these demos, OpenCV was used for its features involving noise reduction, image resizing, and image quality verification. Neva and Elif worked on the ML pipeline development using TensorFlow, Keras, and PyTorch, and selected Python. As part of the investigation, Neva compared candidate ML models for the project's machine learning components, including a basic CNN and transfer learning models such as ResNet50, EfficientNetB0, and MobileNetV2. Samuel worked on the mobile application development using Flutter, SQLite, and Android Studio. Flutter served as the app development framework, being useful to build the navigation screens such as login, create account and home screen; Android Studio was used to establish a connection between the mobile device and the computer, in order to test the application during development; and SQLite served as a lightweight relational database to simply store rows of information that define the stored usernames and passwords in order to login into the app.

Task 2: “Hello World” Demos

The team individually created their starter demos. Sophia implemented a Python demonstration using OpenCV on Google Colab to be able to verify that image quality is not too blurry or unusable, reduce noise from the image, and resize the image to fit the desired dimensions. Neva developed a machine learning demo using Google Colab. She applied and tested three different transfer learning architectures using the pretrained ImageNet weights, including ResNet50, EfficientNetB0, and MobileNetV2, and a basic CNN. Before testing on the selected skin lesion dataset, the CIFAR-10 dataset was used to validate a working pipeline for training, validation, model saving, and evaluation. These experiments were intended to test the working machine-learning pipeline rather than to select a final model for the skin lesion dataset. Samuel worked on the application interface and was able to deploy the Flutter app on a mobile device, with a login screen that includes login and password fields that accept user input. Samuel also created a “create account” button that navigates to a different page and accepts user input to store valid credentials in a lightweight database. After logging in with credentials stored in the database, the app navigates to a home screen where the user can access their track history and capture images.

Task 3: Resolve Technical Challenges

The team investigated the main technical challenges related to image processing, selecting and evaluating machine learning models, and integrating the learned model into the mobile

application. Sophia investigated image-preparation methods to address variations in image size, quality, and noise using Python and OpenCV. Neva and Elif tested CNN and transfer learning techniques and examined suitable evaluation metrics to explore issues in machine learning model selection and performance evaluation. Prior to training on the chosen skin lesion dataset, preliminary tests were conducted to validate the model training and evaluation pipeline. Samuel examined the difficulties of integrating the mobile application with the machine learning model, and chose an effective framework (Flutter) and development environment (Android Studio) that supports the communication between the mobile device and computer. The SQLite database serves as a temporary solution for storing user account information and enabling user authentication. However, a cloud-based storage solution may be necessary in the future to allow the application to communicate with and exchange data with the machine learning model.

Task 4: Compare and Select Collaboration Tools

The team compared and selected collaboration tools regarding software development, communication, task management, presentations, and machine learning development. GitHub was selected as the code exchange, version control, and software development tool. The team chose to coordinate task management and distribution through class discussions and a WhatsApp group chat. Google Slides was selected as the collaborative tool for presentation preparation, while Google Docs was selected for collaborative documentation. Additionally, Google Colab and AI Panther were selected to develop a machine learning model, training processes, and computational resources. The team will utilize these tools and collaboration methods for organization, software development, and machine learning development throughout the project.

Task 5: Requirements Document

The team worked collaboratively to complete the Software Requirements Specification Document. Each member tackled one section of this document; Neva created the document and did the introduction section. Elif was able to complete the section involving the overall description of the system. Samuel completed the specific requirements section, and Sophia completed the section involving additional information.

Task 6: Design Document

To complete the design document, the team worked together to complete 12 sections, with each group member choosing three sections to complete, and then the group went through and provided their feedback. Neva created the document and worked on sections 1, 4, and 5. Her sections involved the introduction, Data Design, and Component Design, respectively. Samuel completed sections 6, Human Interface Design; 8, Security and Privacy Design; and 9, Performance and Optimization. Sophia worked on and completed sections 2, 3, and 12. These sections included System Overview, System Architecture, and Appendices, respectively. Elif completed sections 7 Algorithm and Communication Design, 10 Requirements Traceability Matrix, and 11 References.

Task 7: Test Plan

The team collaboratively wrote the Test Plan based on the system's functional and non-functional requirements. Test cases were developed for functional requirements, including image input and preprocessing, machine learning model training and evaluation, skin lesion prediction, confidence visualization, safety information, history and lesion tracking, data management, and the application interface. Additionally, the Test Plan included test cases for non-functional requirements, such as performance, scalability, accuracy, reliability, usability, security, and maintainability. To ensure the testing process detects potential errors and behaves as expected, both typical and atypical scenarios were considered when developing each test case. As the system develops and additional test cases are required, the Test Plan will continue to be updated to reflect the project's current status.

Contribution of Each Member to the Current Milestone

Samuel Borges: Samuel was entirely responsible for designing the application interface utilizing Flutter, Android Studio, SQLite, and Dart. For creating the first demo, the first step involved the installation of Flutter and Android Studio, with the required Android SDKs that enable communication between the android device and computer. The mobile device selected for testing was a Samsung Galaxy S10+, which successfully built a simple Flutter app on the mobile device. Development of the application then continued by creating the login interface using Dart and organizing the application into separate screen files, including the login screen, create account screen, and home screen. A SQLite database was created within a designated database file, and the account creation process was successfully tested by storing the account ID, username, and password as rows in a users table.

Elif Dila Bayrak: During Milestone 1, Elif mainly focused on machine learning model evaluation. She researched evaluation methods that can be used to measure and compare the performance of different machine learning models. She developed a demo of model evaluation in Google Colab using Python and Scikit-Learn to calculate and display evaluation metrics such as accuracy, precision, recall, F1 score and a confusion matrix. Preliminary testing was performed using the CIFAR-10 dataset to verify that the evaluation pipeline worked correctly before applying it to the skin lesion dataset. The basic CNN achieved a 67.37% test accuracy with an average precision of 0.68, average recall of 0.67 and average F1 score of 0.67. The confusion matrix and training results were also used to identify differences in performance between classes and compare training and validation accuracy. Elif also worked with Neva on exploring model selection and evaluation methods. In addition, she contributed to the Requirements Document, Software Design Document, Test Plan and Progress Evaluation.

Neva Ildem: Neva focused mostly on the project's machine learning component during Milestone 1. As part of the model training demonstration, she investigated and compared suggested CNN and transfer learning techniques, including MobileNetV2, EfficientNetB0,

ResNet50, and a simple CNN. Using ImageNet-pretrained weights, she implemented and trained three transfer learning models as an initial experiment. Preliminary tests were conducted on CIFAR-10 to validate the training and evaluation pipeline before proceeding to the chosen skin lesion dataset. Test accuracy was 37.50% for ResNet50, 13.60% for EfficientNetB0, 66.08% for basic CNN, and 9.67% for MobileNetV2. Since final model comparison and selection would be carried out using skin lesion data in a later milestone, these experiments were utilized to validate the machine learning workflow rather than to choose the final model. To help select suitable candidate models and assessment techniques, Neva also contributed to the literature review of recent skin lesion classification studies. She helped select Google Colab, GitHub, and AI Panther for collaboration and development, which the group will use collectively for computational resources, code exchange, model creation, and organization. She also helped prepare the Milestone 1 presentation, the Requirements Document, the Software Design Document, and the Test Plan.

Sophia Zubieta:

During this milestone, Sophia focused on image preprocessing. She developed an interactive Google Colab demonstration script in Python using OpenCV that identifies images as too blurry and prompts the user to retake the image using Laplacian variance scoring. This demo is also able to reduce noise and uses Lanczos to resize images to the desired 224 x 224 x 3 size. In addition, sharpening of the shape and outline of skin lesion images was achieved along with float tensor normalization ([0.0, 1.0]). These trials were preliminary and would help with future milestones and decision-making. She also contributed to the Milestone 1 presentation, Requirements Document, Software Design Document, Test Plan, and Progress Evaluation.

Plan for Milestone 2

Milestone 2 Task Matrix

Task	Elif	Neva	Samuel	Sophia
Dataset & Preprocessing Pipeline Execution	0%	50%	0%	50%
CNN Model Training	50%	50%	0%	0%
Mobile App Frontend UI	0%	0%	100%	0%
REST API Backend &	0%	0%	0%	100%

Preprocessing Endpoint				
Model Evaluation & Confusion Matrix	50%	50%	0%	0%
App to Server Image Upload	50%	0%	50%	0%

Discussion of Planned Tasks for Milestone 2

Task 1: Dataset & Preprocessing Pipeline Execution:

During Milestone 2, Neva and Sophia will work on preparing the selected unprepared skin lesion dataset for model training. The images will go through preprocessing, which includes resizing, normalization, image quality checking, and other necessary preparation. The goal is to create a consistent and easy-to-use dataset so machine learning models can be evaluated using this dataset.

Task 2: CNN Model Training:

Elif and Neva will work together on training CNN models using the prepared dataset, and different approaches will be tested and compared during the training. The goal for this task will be to develop trained models that can classify skin lesion images as benign or malignant.

Task 3: Mobile App Frontend UI:

Samuel will develop the main user interface for the Android mobile application. This will include the screens and controls needed for the users to navigate the application, select to capture an image, submit it for analysis, and view the returned prediction and confidence score.

Task 4: REST API Backend & Preprocessing Endpoint:

Sophia will work on developing the REST API backend that connects the mobile application to the image processing and machine learning components. The API will receive images submitted by the mobile application and allow them to be prepared for analysis before being sent to the machine learning model.

Task 5: Model Evaluation & Confusion Matrix:

Elif and Neva will evaluate the trained models to compare their performance. Evaluation metrics such as accuracy, precision, recall, F1 score, and confusion matrices will be used. The results will be used. The results will help the team understand model performance and determine which models should continue to be tested and improved.

Task 6: App to Server Image Upload:

Samuel and Elif will work on connecting the mobile application to the server so that images can be uploaded for analysis. The goal is to verify that an image selected or captured through the application can be successfully sent to the server and received for further processing.

Faculty Advisor Meetings

Date(s) of meeting(s) with Client during the current milestone:

- Monday, September 21st, 2026.

Faculty Advisor Feedback on the Current Milestone

- See faculty advisor feedback below.

Task 1 - Compare and Select Technical Tools:

Task 2 - “Hello World” Demos:

Task 3 - Resolve Technical Challenges:

Task 4 - Compare and Select Collaboration Tools:

Task 5 - Requirements Document:

Task 6 - Design Document:

Task 7 - Test Plan:

Faculty Advisor Signature: _____ Date: _____

Evaluation by Faculty Advisor

Faculty Advisor: Please detach and return this page to Dr. Chan or email the scores to pkc@cs.fit.edu.

Elif	1	2	3	4	5	6	6.5	7	7.5	8	8.5	9	9.5	10
Neva	1	2	3	4	5	6	6.5	7	7.5	8	8.5	9	9.5	10

Samuel	1	2	3	4	5	6	6.5	7	7.5	8	8.5	9	9.5	10
Sophia	1	2	3	4	5	6	6.5	7	7.5	8	8.5	9	9.5	10

Faculty Advisor Signature: _____ Date: _____